A direct quartic for Con(ZFC) in 234 integer variables

Exact elimination along common quadratic directions
and cofinal quadratic spacing

AI-assisted mathematical research
Prepared for Vladimir Reshetnikov and the ProveIt project

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Abstract

We give an explicit polynomial $P_{234} \in \mathbb{Z}[x_0, \ldots, x_{233}]$ whose integer unsolvability is provably equivalent in ZFC to Con(ZFC), by composing a new exact reduction with the pinned ProveIt numerical-calculus calibration. It is a sum of 171 explicitly listed quadratic squares, has degree four, 19,050 collected monomials, and coefficient height 4,480. Its actual parent has 240 variables, 177 residuals, 21,482 monomials, and height 6,144. Thus all four size measures improve while degree stays four.

The main reduction identifies six integral directions annihilated by every quadratic Hessian. A unit linear slope permits deletion of one variable and one residual, with an explicit integer polynomial inverse on zero sets. An independently implemented checker verifies all six transformations and every coefficient of the final polynomial. A separate arithmetic change replaces two beta-spacing equations by $y = F^2 + u^2 + v^2$. An elementary prime-power and CRT argument makes this family cofinal among suitable factorial multiples. This produces factorial-free companions in 235 and 236 integer variables.

The contribution is a concrete reduction of the direct small-coefficient branch, not a global minimum or a new universal-variable bound. The algebra and cofinality arguments are ordinary unbounded proofs; finite coefficient verification and the inherited logical calibration have distinct scopes. No end-to-end Lean or Rocq proof is claimed.

Comparison snapshot: `7a8da414376a80b90ecf799f37ceb26fb9ac674e`.
All final variables range over the unrestricted integers.
Every principal residual is printed in Appendix A.

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1 The explicit result

The principal mathematical object is the fixed ordinary polynomial

$$\boxed{P_{234}(x_0,\dots,x_{233}) = \sum_{i=0}^{170} \mathcal{Q}_i(x_0,\dots,x_{233})^2.} \tag{1}$$

Appendix A specifies every coefficient of every $\mathcal{Q}_i$. The file `equations/P234/equation.json` is exactly the same definition in sparse machine-readable form. Its neighboring `expanded_polynomial.json` lists all 19,050 collected terms of P_{234} directly. Neither definition contains a variable exponent, binomial coefficient, proof-checking oracle, unspecified universal polynomial, or uncomputed program-dependent integer.

For orientation, the first six residuals are

$$\begin{aligned}
\mathcal{Q}_0 &= x_4 - x_3^2,\\
\mathcal{Q}_1 &= x_0 - x_0x_2 - x_2 + x_9,\\
\mathcal{Q}_2 &= -x_2x_7 - x_3 + x_8x_9,\\
\mathcal{Q}_3 &= -1 + x_1 - x_8x_9,\\
\mathcal{Q}_4 &= 1 + x_4 - x_5x_{20} + x_{18},\\
\mathcal{Q}_5 &= -1 - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{21}.
\end{aligned}$$

The other 165 residuals are part of the definition, not omitted hypotheses. Every residual coefficient has absolute value at most 64. After squaring and collecting, the coefficient of x_0^4 is 13, so the degree is exactly four. The largest absolute coefficient is attained, for example, at $x_{11}^2x_{22}^2$, whose coefficient is 4,480.

Theorem 1.1 (Principal consistency equation)**.** *For the polynomial* (1)*, the new algebraic proof composed with the pinned ProveIt calibration establishes*

$$\mathrm{ZFC} \vdash \big(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{234} \quad P_{234}(\boldsymbol{x}) = 0\big) \iff \mathrm{Con(ZFC)}.$$

More specifically, ZFC proves an explicit integer zero-set bijection between the new residual system and the pinned 240-variable parent. No consistency assumption is used to establish this reduction.

The exact new result proved independently of the inherited semantics is

$$\mathrm{ZFC} \vdash \big(\exists \boldsymbol{x} \in \mathbb{Z}^{234}\ P_{234}(\boldsymbol{x}) = 0\big) \iff \big(\exists \boldsymbol{z} \in \mathbb{Z}^{240}\ P_{240}(\boldsymbol{z}) = 0\big). \tag{2}$$

The parent theorem identifies the latter existence statement with finite syntactic ZFC refutability. Section 2 states that dependency precisely; it is not silently reclassified as a new formal verification.

1.1 Measured comparison and the objective

Equation	Variables	Residuals	Degree	Monomials	Height
Shared-error parent P_{240}	240	177	4	21,482	6,144
Principal P_{234}	**234**	**171**	**4**	**19,050**	**4,480**
Factorial-free parent	242	178	4	5,018	7,968
Factorial-free $P_{235,\beta}$	235	170	4	18,550	4,576
Smaller-support $P_{236,\beta}$	236	171	4	4,947	16,866

Height means the largest absolute coefficient of the fully collected polynomial, rather than of an uncollected expression or of one residual. Monomials are counted after all cancellations. The three delivered outputs have different advantages: P_{234} has the smallest dimension and height; $P_{235,\beta}$ uses the factorial-free arithmetic source and has one fewer residual; $P_{236,\beta}$ has substantially smaller support.

Relative to its actual parent, P_{234} removes six variables and six residuals, 2,432 collected monomials, and 1,664 from the coefficient height. The reductions are 2.5%, approximately 11.32%, and approximately 27.08% for

variables, monomials, and height, respectively. This is a concrete improvement of that branch, not a major new universal bound.

The inspected repository already contains nine-positive-variable and other low-variable constructions, including quartics in 37 unrestricted integers on a different branch [2]. Consequently 234 is not a project-wide variable record. The present objective is a direct numerical proof-calculus interpretation, degree four, small literal coefficients, and an auditable finite definition. No scalar measure called "simplest" orders all these resources. The distinction between degree, variable count and explicit representation is also central in current work on universal Diophantine bounds [5, 6].

2 What the parent certifies

The comparison source is the unchanged incoming archive *Shared Factorial Error*, at the pinned commit [3, 1]. Its selected file is `equations/equation240/equation.json`. The archive hash is recorded in `SOURCES.json`, and the exact member bytes are checked against the bundled input on every release verification.

The parent uses the numerical calculus C_{222}, with truth code 0 and falsity code 1. It represents a finite multiset of single-conclusion Horn rule instances together with a balance condition. Bounded row certificates and a Chinese remainder construction enforce the local arithmetic. A shared Pell construction represents the required powers by ordinary polynomial equations. The parent does not use a simulation of K215 or a specialization of Jones's universal polynomial in its emitted definition.

The logical endpoints inherited from ProveIt are

`KEndpoint.c222_iff_zf_refutation`,
`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

They identify finite C_{222} refutability with syntactic ZF refutability and then syntactic ZFC refutability. The ZF/ZFC step uses the established constructible-universe equiconsistency argument. The retained source archives identify the numerical rules and ordinary arithmetic-to-calculus proofs as well as these logical interfaces.

A useful reason that the multiset representation is sound is the following finite closure argument. Pad unused premise slots by derivable truth. Suppose at every positive label the number of conclusions minus premises is zero, except at the target label where it is one. If the target were outside the finite Horn closure of truth, every rule with an outside conclusion would have at least one outside premise. Summing conclusions minus premises over the outside labels would give a number at most zero, contradicting one. Thus circular collections cannot manufacture a false refutation merely by balancing multiplicities.

In the shared-error source, the arithmetic quantity F is authenticated as $t!$ by a nearest-integer fixed point. The normalized common error supplies

$$\left| F - \frac{(F^2+1)^t}{\binom{F^2+t}{t}} \right| < \frac{1}{2}.$$

For $t \geq 16$ and the signs recovered in the parent, its unbounded proof forces $F = t!$. A shared norm simultaneously controls this error and eleven digit remainders. The signed-base power lemma is used before signs of the radix and binomial output are known. These ordering issues matter: a positivity conclusion obtained from a decoded digit cannot be assumed in the estimate that first decodes that digit.

The independent ordinary review in `proofs/parent_arithmetic_review.md` checks the changed signed bootstrap, the separate unit-base and $F = 0$ cases, the nearest-factorial estimate, and the compatible witness construction. It found no material gap in those arguments under their retained Pell and calibration interfaces. The original release verifier was also rerun successfully, including all 242 sealed files, its earlier source transformations, the 240-variable profile, and its printed equation.

These checks do not constitute a fresh build of the entire transitive Lean or Rocq development. The new contribution proves (2) by elementary algebra, and then composes with the identified ordinary parent theorem. This explicit dependency is the mathematical scope of Theorem 1.1.

3 Elimination along a common quadratic direction

Write an integer quadratic polynomial as

$$Q_i(\boldsymbol{x}) = q_i(\boldsymbol{x}) + \ell_i(\boldsymbol{x}) + c_i,$$

where q_i is homogeneous of degree two, ℓ_i is linear, and c_i is constant. Let H_i be its integer Hessian: the diagonal entry for a term ax_j^2 is $2a$, and a mixed term bx_jx_k contributes b at each of the two symmetric off-diagonal positions.

Lemma 3.1 (Constant directional slope)**.** *If $\boldsymbol{v} \in \mathbb{Z}^n$ satisfies $H_i\boldsymbol{v} = 0$, then for all $\boldsymbol{y} \in \mathbb{Z}^n$ and $s \in \mathbb{Z}$,*

$$Q_i(\boldsymbol{y} + s\boldsymbol{v}) = Q_i(\boldsymbol{y}) + s\,\ell_i(\boldsymbol{v}).$$

Proof. Expanding the homogeneous quadratic part gives the cross term $s\boldsymbol{y}^T H_i\boldsymbol{v}$ and the pure term $s^2 q_i(\boldsymbol{v})$. The former vanishes. Since $2q_i(\boldsymbol{v}) = \boldsymbol{v}^T H_i\boldsymbol{v} = 0$, the latter vanishes as well. The linear and constant parts give the stated formula. □

Theorem 3.2 (Integral quadratic projection)**.** *Let $Q_0, \ldots, Q_{m-1} \in \mathbb{Z}[x_0, \ldots, x_{n-1}]$ have degree at most two. Suppose there are $\boldsymbol{v} \in \mathbb{Z}^n$ and indices p, r such that*

$$H_i\boldsymbol{v} = 0 \quad \text{for every } i, \qquad v_p = 1, \qquad a_r := \ell_r(\boldsymbol{v}) \in \{-1, 1\}.$$

Set $a_i = \ell_i(\boldsymbol{v})$. Regard $\boldsymbol{y}$ as a vector with $y_p = 0$ and $n-1$ free integer coordinates, and define

$$R_i(\boldsymbol{y}) = Q_i(\boldsymbol{y}) - a_i a_r Q_r(\boldsymbol{y}), \qquad i \neq r. \tag{3}$$

These are integer quadratics. Their common integer zero set is in explicit bijection with the common integer zero set of the original Q_i.

Proof. Given a new zero, put

$$s = -a_r Q_r(\boldsymbol{y}), \qquad \boldsymbol{x} = \boldsymbol{y} + s\boldsymbol{v}. \tag{4}$$

Every coordinate is an integer. Lemma 3.1 gives $Q_r(\boldsymbol{x}) = Q_r(\boldsymbol{y}) + sa_r = 0$, because $a_r^2 = 1$. For every other row it gives $Q_i(\boldsymbol{x}) = R_i(\boldsymbol{y}) = 0$.

Conversely, given an old zero, let $s = x_p$ and $\boldsymbol{y} = \boldsymbol{x} - x_p\boldsymbol{v}$. Then $y_p = 0$. The same identity gives $Q_r(\boldsymbol{y}) = -x_p a_r$, so the scalar reconstructed by (4) is exactly x_p. For every $i \neq r$,

$$R_i(\boldsymbol{y}) = -x_p a_i - a_i a_r(-x_p a_r) = 0.$$

The two constructions are inverse. Formula (3) is a constant integer row combination restricted to $y_p = 0$, hence has degree at most two. There is no division or residual congruence condition. □

Corollary 3.3. *Replacing $\sum_i Q_i^2 = 0$ by $\sum_{i \neq r} R_i^2 = 0$ removes one integer unknown and one quadratic residual without increasing degree above four. The equivalence is provable by finite arithmetic in ZFC.*

Proof. A sum of integer squares is zero precisely when each summand is zero. Apply Theorem 3.2. □

3.1 A small example

Consider the residuals

$$u - a^2, \qquad v - b^2, \qquad w - (u+v)c.$$

The direction that increases u and decreases v annihilates all three quadratic Hessians. With $s = u + v$, the system becomes

$$s - a^2 - b^2 = 0, \qquad w - sc = 0.$$

It has one fewer integer variable and one fewer residual. A zero of the new system restores $u = a^2$, $v = b^2$, and the inverse uses $s = u + v$. A procedure that only substitutes a single existing variable will initially see cubic terms in uc or vc. Changing to the combination $u + v$ exposes the quadratic-preserving reduction.

3.2 Integrality is an essential hypothesis

A primitive integral direction alone does not justify division by its slope. If all slopes have a common factor greater than one, solving for the eliminated coordinate can leave a divisibility condition. Our certificates check $v_p = 1$ and a literal slope ± 1 at every step. More general primitive directions and slope vectors can be handled by unimodular completions and Bezout combinations, but that generalization is not needed for any claimed count in this release.

The proof also uses the unrestricted integer domain. An invertible integer change of coordinates need not preserve positivity or nonnegativity. The present reduction is not automatically a reduction of the repository's nine-positive-variable equations.

4 The six eliminations and the coefficient reduction

The discovery program stacks the sparse Hessian rows and computes their common kernel by exact rational Gaussian elimination. It clears denominators, makes each vector primitive, and considers coordinates with unit entries. It then checks every Hessian product and every candidate unit slope exactly. The selected coordinate change and unit substitution are stored as a finite certificate.

For the principal parent, the six deleted register names, in order, are

Step	Input variables	Output variables	Output residuals	Deleted register
1	240	239	176	`add.71`
2	239	238	175	`mul.188`
3	238	237	174	`mul.303`
4	237	236	173	`sqd.11`
5	236	235	172	`add.277`
6	235	234	171	`add.112`

These are historical wire names, not mathematical operations remaining in the final polynomial. The coordinate indices, complete integer direction vectors, selected residuals, unit slopes and restoration polynomials are specified in `stages/radical234/projection_certificate.json`. Renumbering after each deletion is recorded explicitly.

Immediately after these six steps the quartic has 20,739 monomials and height 18,542. We then apply 147 recorded elementary operations

$$R_i \longleftarrow R_i + kR_j, \qquad i \neq j, \quad k \in \mathbb{Z}. \tag{5}$$

Each is invertible by subtracting kR_j. Thus the common zero set is unchanged, although the polynomial obtained by summing squares usually changes. The operations give the final 19,050 monomials and height 4,480. They are recorded in `equations/P234/row_certificate.json`.

The row search chooses reductions of the squared Euclidean norm of the coefficient vectors. Its heuristic has no role in the correctness proof. The checker replays the frozen operations exactly and then replays them backwards, recovering the original intermediate rows. This is the reason that a changed optimization heuristic need not be trusted or rerun.

The independent projection checker uses (3) directly. The constructor uses a shear followed by substitution. These two implementations agree coefficient by coefficient on every intermediate residual. Numerical evaluations at arbitrary small tuples supplement these identities as regression controls; they are not substitutes for the identities and are not assumed zeros of the consistency equation.

5 A second contribution: cofinal quadratic spacing

The factorial-free parent [4] uses

$$t = 16 + h^2, \qquad F = [2(t+1)]^t,$$

where the displayed power is represented by the inherited ordinary Pell certificate. Its beta spacing is specified by two quadratic register rows

$$m = (2w+1)^2, \qquad y = Fm. \tag{6}$$

The spacing must be large enough to encode a finite accumulator path; in the converse construction it is chosen as a multiple of $t!$. Authenticating $F = t!$ is not part of this branch.

We replace (6) by the single quadratic row

$$\boxed{y = F^2 + u^2 + v^2, \qquad u, v \in \mathbb{Z}.} \tag{7}$$

The already stored value F^2 is reused. This uses the same two witness slots, removes one residual, and implies $y \geq F^2 > F + 1$ once the existing growth certificate has been interpreted. The following theorem supplies exactly the completeness property needed.

Lemma 5.1 (Two squares modulo an odd prime power)**.** *For every odd prime p and $e \geq 1$, there exist integers a, b with $a^2 + b^2 \equiv -1 \pmod{p^e}$.*

Proof. The squares in $\mathbb{F}_p$ and minus one minus those squares are two sets of size $(p+1)/2$, so they intersect. A resulting solution modulo p cannot have both coordinates zero. Suppose the first is nonzero; the other case is symmetric. Given a solution modulo p^j, change a to $a + \delta p^j$, where

$$\delta \equiv -\frac{1 + a^2 + b^2}{p^j}(2a)^{-1} \pmod{p}.$$

Expansion proves the congruence modulo p^{j+1}. The coordinate remains nonzero modulo p, so induction completes the lift. □

Theorem 5.2 (Cofinal factorial multiples)**.** *Let $t \geq 1$, $F \geq 2$, and assume $2^{v_2(t!)} \mid F^2$. For every integer bound C, there are integers u, v such that*

$$y = F^2 + u^2 + v^2 > C, \qquad t! \mid y.$$

Both u and v can be chosen as multiples of F. If F is even, the resulting y is even.

Proof. Write $t! = 2^a m$ with m odd. Lemma 5.1 and the Chinese remainder theorem give $m \mid 1 + A^2 + B^2$. This also holds when $m = 1$, with the congruence vacuous. For positive integers k, put

$$u_k = F(A + km), \qquad v_k = FB, \qquad y_k = F^2\big(1 + (A + km)^2 + B^2\big).$$

The odd part m divides y_k, and 2^a divides F^2. Hence $t!$ divides y_k. The family is unbounded, so some y_k exceeds C. Its parity when F is even is immediate. □

For the actual $F = [2(t+1)]^t$, we have $v_2(F) \geq t > v_2(t!)$, so the hypothesis holds with a large margin. The theorem is elementary and unbounded. The delivered program constructs 195 exact examples for $t = 16, \ldots, 80$ and three offsets each; those examples are checks of the constructor, not the proof of cofinality.

5.1 Soundness in the required order

The known positive horizon and positive base first recover $F = [2(t+1)]^t$ from the unchanged power certificate. Only then does (7) imply $y \geq F^2 > F + 1$. The retained radix is an odd square $q = (2z+1)^2$, hence at least one. Its numerator power requests at base $q + 1 \geq 2$ make the nine actual CRT representatives nonnegative; therefore their offset sum H is at least two.

The retained index row has the form $B\iota = C_0 + F^2 + 1$, with a nonnegative representative C_0, and excludes $B = 0$. Thus $|BH| \geq 2$. The signed-power lemma gives $x \geq 0$ and $q = (BH)^x$. The endpoint

$$x = 1 + [1 + (t+1)y]u_t$$

excludes $x = 0$, since its bracket is greater than one and cannot divide -1. In the other endpoint,

$$x = F + 1 + (y+1)u_0,$$

a negative integer u_0 would force $x \leq F - y < 0$. Hence

$$u_0 \geq 0, \qquad x \geq F + 1, \qquad u_t \geq 1, \qquad x \geq 2 + (t+1)y \geq 2 + (t+1)F^2.$$

These bounds strengthen the parent bounds. Its centered-remainder proof can therefore recover the positive binomial value $B = \binom{F^2+t}{t}$, and its generalized CRT theorem recovers every bounded local rule. No sign of that binomial output is assumed before its digit is authenticated.

The initial accumulator $F+1$ is now a proper remainder modulo $y+1$ immediately. The old boundary case $y = F$, which required a zero-path contradiction, is impossible. The same positive-weight product and finite Horn-closure argument yield a C_{222} refutation.

5.2 Completeness and the parity choices

Start with a finite C_{222} refutation. Choose a sufficiently large permitted horizon $t = 16+h^2$, form F, and construct its finite positive accumulator path before choosing y. Theorem 5.2 supplies an even multiple of $t!$ exceeding every accumulator and $F+1$. The moduli $1+(i+1)y$ for $0 \le i \le t$ are pairwise coprime: a common prime divides the index difference, hence is at most t and divides y, contrary to its dividing $1+(i+1)y$. They are all odd.

CRT gives the required beta code x. Adding their odd product changes parity, so x can be chosen even and arbitrarily large. Define the same B and $N = B^x$ as in the parent and use its generalized binomial CRT at every prime of B. The value B is odd because $v_2(F^2) > \lfloor \log_2 t \rfloor$: each F^2+i has the same two-adic valuation as i, so those valuations cancel in the binomial quotient. The precision modulus is therefore odd and its representative classes have free parity. The parent's choices of a dominating even anchor and odd H remain available. Then $q = (BH)^x$ is an odd square.

The other power, remainder, and norm witnesses are supplied by the unchanged constructive arguments. Thus the new spacing preserves the existence equivalence with a finite refutation. It generally does *not* extend an individual zero to the old block (6): y/F need not be an odd square. We use an existential soundness/completeness proof for this step, rather than claiming a coordinate bijection.

6 The two factorial-free companions

In the parent's final 242-variable chart, the actual spacing is $x_2 - 1$, $F = x_3$, and its existing square is x_4. The two private witness slots are 240 and 241. The literal source change is

$$\begin{aligned} -1 + x_2 - x_3 x_{240} \quad &\rightsquigarrow \quad -1 + x_2 - x_4 - x_{240}^2 - x_{241}^2, \\ -1 + x_{240} - 4x_{241} - 4x_{241}^2 \quad &\text{is deleted.} \end{aligned}$$

The row $x_4 - x_3^2$ is retained. The checker verifies that neither private witness occurs anywhere else. The old projection and row certificates also leave this block untouched; the arithmetic review records this transport explicitly.

The new source has 242 variables and 177 residuals. Six applications of Theorem 3.2 give 236 variables and 171 residuals. One further elementary row operation gives $P_{236,\beta}$, with 4,947 monomials and height 16,866. The definition is

$$P_{236,\beta} = \sum_{i=0}^{170} S_i^2,$$

where every S_i is listed literally in `equations/P236_beta/equation.json`.

For the 235-variable companion, an affine residual in the 236-variable intermediate chart permits the following exact substitution:

$$\begin{aligned} x_{96} = {} & x_9 + x_{10} + 4x_{20} + 2x_{23} + 2x_{27} + 2x_{29} + 2x_{31} \\ & + 2x_{33} + 4x_{35} + 2x_{36} + 2x_{39} + 2x_{42} + 4x_{61} + 2x_{62}. \end{aligned}$$

Its unit coefficient makes the restoration integral and unique. Deleting that row and substituting everywhere leaves 235 variables and 170 residuals, still quadratic. After 433 reversible integer row operations, the resulting quartic $P_{235,\beta}$ has 18,550 monomials and height 4,576. All its coefficients appear in `equations/P235_beta/equation.json` and its expanded neighbor.

Corollary 6.1. *The ordinary new proofs, composed with the factorial-free parent's calibration, give in ZFC both equivalences*

$$\neg\exists \boldsymbol{x} \in \mathbb{Z}^{235}\ P_{235,\beta}(\boldsymbol{x}) = 0 \iff \mathrm{Con(ZFC)}, \qquad \neg\exists \boldsymbol{x} \in \mathbb{Z}^{236}\ P_{236,\beta}(\boldsymbol{x}) = 0 \iff \mathrm{Con(ZFC)}.$$

7 Verification and its limits

7.1 Exact finite statements that were checked

The independent checker is a standard-library Python program and imports no discovery or projection implementation. Its checks include:

1. Both unchanged source-archive hashes and the exact selected equation members; every intermediate certificate is tied to its source.
2. Every Hessian-kernel identity, unit direction coordinate, unit slope, deleted residual, renumbering, and restoration polynomial.
3. The direct identity (3), independently of the producer's shear-and-substitution algorithm.
4. The extra affine substitution by the independent identity $Ax_p^2 + Bx_p + C \mapsto AS^2 + BS + C$.
5. Every elementary integer row operation and its reverse replay.
6. All coefficients of each complete expanded sum of squares, and all dimensions, residual counts, degrees, supports, and heights.
7. A separate parser for the printed principal residuals, so the appendix is checked against the literal coefficient data.

The projection checker rejects eight kinds of deliberate corruptions; the affine checker rejects five, and the row checker rejects four. Representative checks also passed with Python optimization enabled, so correctness gates do not depend on removable `assert` statements. The full package verification is reproducible with `python3 -B verify.py`.

These finite checks establish the supplied algebraic certificates. The cofinality theorem and the inherited arithmetic/logical calibration are unbounded mathematical arguments. The release does not call samples an unbounded proof or call a Python program a proof-assistant kernel.

7.2 Two useful but limited rank facts

For the principal system, the stacked Hessian has full column rank 234 modulo the explicitly verified prime 1,000,003. A nonzero minor modulo a prime is nonzero over $\mathbb{Q}$, so its common Hessian radical over $\mathbb{Q}$ is zero. Thus no further nonzero constant direction makes *every* residual have a coordinate-independent slope.

Separately, the homogeneous quadratic parts of its 171 residuals have row rank 171 over $\mathbb{Q}$. Hence no nonzero constant linear combination of the residuals is affine. The 235-variable companion has the corresponding full row rank 170. These facts certify exhaustion of the two particular linear-algebra moves on the displayed presentations.

They do not establish a minimum under arbitrary transformations. For a small counterexample, take

$$R_1 = x^2 - 1, \qquad R_2 = xy - 1, \qquad R_3 = y^2 - 1.$$

Their quadratic parts are independent and their common Hessian radical is zero. Nevertheless,

$$yR_1 - xR_2 = x - y.$$

This polynomial-multiplier identity supplies $x = y$, reducing the system to $x^2 - 1 = 0$. Both rank checks hold before that reduction. Nonlinear coordinate changes, polynomial residual combinations, and a different semantic encoding therefore remain outside the limited certificate.

8 Further research questions

Polynomial residual combinations. Can a bounded degree-three Macaulay computation find a useful affine consequence of the 234-variable system? The generators would be its residuals and their products with single

variables. A successful candidate should carry an explicit polynomial-combination identity and an integral restoration map. The counterexample above shows why this question is strictly stronger than either rank test already completed.

Integral lattice generalization. The current implementation uses directions with a unit coordinate and a single unit residual slope. Can Hermite or Smith normal form expose more useful choices earlier in the source pipeline? Any nonunit invariant factor must be retained as a divisibility condition. A rational elimination without that accounting is insufficient for integer solutions.

Joint source and coordinate optimization. The same six common directions occur in several recent direct-source variants. Can the source compiler use shared combinations as registers from the outset? Doing so could avoid the later coefficient growth, and could permit a lower-support version of the 234-variable endpoint. This should be measured against the entire coefficient list, not a partial gate count.

A smaller cofinal arithmetic family. The family $F^2 + u^2 + v^2$ handles every odd prime power and uses the already available two-adic valuation of F. Is there a one-witness family with the same uniform divisibility and cofinality properties? The tempting family $F^2 + u^2$ cannot generally replace it: if a prime $p \equiv 3 \pmod 4$ does not divide F, the congruence $u^2 \equiv -F^2 \pmod p$ is impossible. Any proposed one-witness replacement must explain how it avoids this local obstruction.

Formalizing the short new interface. The integral projection theorem is a compact first target: its proof needs only polynomial evaluation, the constant directional-slope identity, and integer unit arithmetic. The cofinal-spacing theorem is another contained target, using finite-field square counts, elementary lifting, and CRT. These are better delimited milestones than treating successful finite coefficient checks as a completed internal ZFC development.

9 Files and integration

The archive includes this LaTeX source and PDF, the complete principal residual appendix, all three final coefficient lists and expansions, exact intermediate data, construction and independent verification code, ordinary proof reviews, immutable source archives, and a checksum manifest. Its intended intake location is `docs/incoming`, followed by an additive placement in the DiophantineConsistency collection. No existing repository file was changed.

A polynomial term is encoded as `[coefficient,[variable indices]]`; repeated indices represent repeated factors and an empty index list is constant. The polynomial is the sum of the squares of its residuals. Variable names are explanatory metadata. The arrays and their integer coefficients are the definition.

The new checks need only Python's standard library and no network. A user can evaluate a supplied tuple with `code/evaluate.py`. The verification scripts do not attempt to locate a refutation or a polynomial zero. Run `pdflatex` twice on `report/report.tex` to rebuild the PDF; typesetting versions may change PDF bytes without changing any mathematical object.

The proof dependencies are deliberately visible. The result is a new, fully emitted equation and a checked reduction from its documented source. It is not a proof that this equation has no integer solution, nor a claim that the broader search for the simplest consistency equation is finished.

References

[1] V. Reshetnikov and ProveIt contributors, *ProveIt*, repository snapshot `7a8da414376a80b90ecf799f37ceb26fb9ac674e`, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/7a8da414376a80b90ecf799f37ceb26fb9ac674e.

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[3] ProveIt incoming research, *Shared Factorial Error*, 25 October 2026, `ConZFC_Shared_Factorial_Error_2026-10-25.zip`, retained unchanged in `sources/`. Original location: `docs/incoming` at [1]. The selected parent is its `equations/equation240/equation.json`.

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A Every quadratic residual of the principal polynomial

This appendix is the complete definition of the 234-variable equation. All $x_0, \ldots, x_{233}$ are unrestricted integers, and $P_{234} = \sum_{i=0}^{170} \mathcal{Q}_i^2$. Coefficients and repeated variable indices are checked against the JSON definition by a separate parser. Line breaks in a residual have no algebraic meaning.

$$\mathcal{Q}_0 = -x_3^2 + x_4$$

$$\mathcal{Q}_1 = x_0 - x_0x_2 - x_2 + x_9$$

$$\mathcal{Q}_2 = -x_2x_7 - x_3 + x_8x_9$$

$$\mathcal{Q}_3 = -1 + x_1 - x_8x_9$$

$$\mathcal{Q}_4 = 1 + x_4 - x_5x_{20} + x_{18}$$

$$\mathcal{Q}_5 = -1 - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{21}$$

$$\begin{aligned}\mathcal{Q}_6 = {} & -1 + x_0 - x_3 + x_3x_{10} - 6x_{10} \\ & + 3x_{10}^2 - x_{10}x_{11} - x_{10}x_{12} - 8x_{10}x_{21} - 11x_{11} \\ & - 5x_{11}^2 + 7x_{11}x_{12} - 16x_{11}x_{21} + x_{12} - 3x_{12}^2 \\ & - 5x_{21} + 12x_{21}^2 + x_{22} - 2x_{25} + 2x_{26} \\ & - x_{38} + 3x_{73} - x_{78} - x_{208}^2 - x_{209}^2 \\ & - x_{210}^2 - x_{211}^2 - x_{212}^2 + x_{218}^2 + x_{219}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_7 = {} & -1 - x_1 + x_8x_9 + 2x_{10} - x_{10}^2 \\ & - 4x_{10}x_{22} - 4x_{22} - 4x_{22}^2 + x_{23}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_8 = {} & -24x_{10}x_{23} + 24x_{10}x_{35} + 2x_{23} - 4x_{23}^2 + x_{24} \\ & - 2x_{35} + 4x_{35}^2 - x_{36}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_9 = {} & -1 - x_1 + x_8x_9 - x_{11} - x_{11}^2 \\ & - 2x_{11}x_{12} - x_{12} - x_{12}^2 + x_{25}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{10} = {} & -1 + x_{10} - x_{10}^2 - 2x_{10}x_{12} + x_{12} \\ & - x_{12}^2 + x_{26}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{11} = {} & 2 + 4x_0 + x_0x_2 + 2x_1 + x_2 \\ & - x_2x_7 - x_3x_{10} - x_3x_{11} - x_4 + x_5x_{20} \\ & - x_8x_9 - x_9 + 3x_{10} + 7x_{10}^2 - 15x_{10}x_{11} \\ & - 11x_{10}x_{12} - 16x_{10}x_{21} + 8x_{11} + 9x_{11}^2 - 13x_{11}x_{12} \\ & - 8x_{11}x_{25} - 4x_{11}x_{26} - 11x_{12} + 4x_{12}^2 - 2x_{12}x_{13} \\ & - 28x_{12}x_{21} - x_{13}^2 - x_{18} - 13x_{21} + 8x_{21}^2 \\ & - 8x_{21}x_{26} + 2x_{24} - 11x_{25} - 20x_{25}^2 - 8x_{25}x_{26} \\ & + 4x_{26} - 4x_{26}^2 + x_{27} + 2x_{28} + x_{29} \\ & + 2x_{30} - x_{31} - 4x_{31}^2 + 3x_{32} + 2x_{34} \\ & + 4x_{36} + 2x_{37} - x_{38} + 2x_{40} + x_{41} \\ & + 2x_{43} + 2x_{45} + x_{61} - 4x_{62}^2 + 3x_{63} \\ & - 7x_{73} + x_{77} + x_{78} + x_{138}x_{192} + x_{192} \\ & - x_{193} + x_{204}x_{217} - 5x_{208}^2 - x_{212}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{12} = {} & 1 - x_0 + 9x_{10} - 4x_{10}^2 + x_{10}x_{11} \\ & + 3x_{10}x_{12} - 8x_{10}x_{25} + 2x_{11}^2 + 3x_{11}x_{12} - 5x_{12} \\ & + 4x_{12}^2 - 2x_{12}x_{13} - x_{13}^2 + x_{21} + 5x_{25} \\ & - x_{26} - 4x_{27}^2 + x_{28} - 4x_{33} - 4x_{33}^2 \\ & + x_{34} + x_{41} - 2x_{45} - 3x_{73} + x_{208}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{13} = {} & - 8x_{10} - x_{10}x_{11} + x_{10}x_{12} + 8x_{10}x_{25} + 8x_{10}x_{26} \\ & - 6x_{11} + x_{11}^2 + 3x_{11}x_{12} + 8x_{11}x_{25} + 8x_{11}x_{26} \\ & - 4x_{21} + 7x_{25} - 4x_{25}^2 - 8x_{25}x_{26} + 8x_{26} \\ & - 4x_{26}^2 + x_{29} + x_{73}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{14} = {} & - 1 - x_1 + x_3x_{11} + x_8x_9 - x_{10} \\ & + 6x_{10}^2 + 7x_{10}x_{11} + x_{10}x_{12} + 8x_{10}x_{21} - 8x_{10}x_{26} \\ & - 8x_{10}x_{29} + 4x_{11} - 14x_{11}^2 + 11x_{11}x_{12} - 16x_{11}x_{21} \\ & + 8x_{11}x_{25} - 24x_{11}x_{26} - 16x_{11}x_{29} - 4x_{11}x_{49} - x_{12} \\ & + 6x_{12}^2 - 2x_{12}x_{13} + 8x_{12}x_{26} + 4x_{12}x_{29} - 8x_{12}x_{49} \\ & - x_{13}^2 + 3x_{21} - 4x_{21}^2 - 16x_{21}x_{26} - 8x_{21}x_{29} \\ & + 8x_{25} - 4x_{25}^2 - 8x_{25}x_{26} + 15x_{26} - 4x_{26}^2 \\ & - 4x_{27}^2 + x_{28} + 5x_{29} + x_{30} + x_{41} \\ & - x_{44} - 2x_{45} - 4x_{49}^2 + x_{50} - x_{73} \\ & - x_{77}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{15} = {} & - 1 + 2x_0 - x_1 + x_8x_9 + 7x_{10} \\ & + 9x_{10}^2 - 12x_{10}x_{11} - 10x_{10}x_{12} - 12x_{10}x_{25} + 16x_{10}x_{26} \\ & + 24x_{10}x_{39} + 4x_{10}x_{54} - 12x_{11} + 16x_{11}^2 + 2x_{11}x_{12} \\ & - 8x_{11}x_{25} + 4x_{11}x_{26} - 20x_{12} - 2x_{12}^2 - 16x_{12}x_{25} \\ & - 8x_{12}x_{26} + 16x_{12}x_{39} + 8x_{12}x_{54} + 2x_{21} - 19x_{25} \\ & - 12x_{25}^2 - 8x_{25}x_{26} + 3x_{26} - 4x_{26}^2 + 8x_{26}x_{39}\end{aligned}$$

$$
\begin{aligned}
&- 8x_{27}^2 + 2x_{28} + x_{29} + x_{31} + 8x_{33} \\
&+ 8x_{33}^2 - 2x_{34} - 4x_{39} + 4x_{39}^2 - x_{40} \\
&- 3x_{45} + 4x_{54} + 4x_{54}^2 - x_{55} - 16x_{73} \\
&- 2x_{208}^2
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{Q}_{16} = \; &- 2 - x_0 - 2x_1 + x_2x_7 + x_3x_{10} \\
&+ x_3x_{11} + x_4 - x_5x_{20} + x_8x_9 + 9x_{10} \\
&+ 14x_{10}^2 - x_{10}x_{11} + 5x_{10}x_{12} - 8x_{10}x_{25} + 16x_{10}x_{26} \\
&+ 24x_{10}x_{39} + 5x_{11} - x_{11}^2 - x_{11}x_{12} - 8x_{11}x_{25} \\
&- 8x_{11}x_{26} + 6x_{12} + 8x_{12}x_{25} + 16x_{12}x_{26} + 16x_{12}x_{39} \\
&+ x_{18} + 7x_{21} - 2x_{25} + 8x_{25}x_{26} + 5x_{26} \\
&+ 8x_{26}^2 + 8x_{26}x_{39} - x_{29} - 8x_{33} - 8x_{33}^2 \\
&+ 2x_{34} - 4x_{39} + 4x_{39}^2 - x_{40} + x_{45} \\
&- 5x_{73} - x_{77} - x_{78} + x_{208}^2
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{Q}_{17} = \; &- 1 + 2x_0 - x_1 + x_3x_{11} + x_8x_9 \\
&+ x_{10} + 5x_{10}^2 - 2x_{10}x_{11} + 6x_{10}x_{12} + 4x_{10}x_{25} \\
&+ 16x_{10}x_{26} - 5x_{11} + x_{11}^2 - 2x_{11}x_{12} - 16x_{11}x_{25} \\
&+ 4x_{11}x_{26} - 2x_{12} - 2x_{12}^2 + 24x_{12}x_{26} + 2x_{21} \\
&- 14x_{25} - 16x_{25}^2 + 16x_{26} + 4x_{26}^2 + 4x_{27}^2 \\
&- x_{28} - x_{31} - 4x_{31}^2 + x_{32} - 4x_{33} \\
&- 4x_{33}^2 + x_{34} + 5x_{45} - 2x_{73} - x_{77} \\
&- 2x_{208}^2 - x_{211}^2 - x_{212}^2 + x_{219}^2
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{Q}_{18} = 1 \; &- 2x_{10} + 4x_{10}^2 + 4x_{10}x_{11} - 4x_{10}x_{12} \\
&- 8x_{10}x_{26} - 8x_{10}x_{61} - 20x_{11} - 7x_{11}^2 + 14x_{11}x_{12} \\
&+ 8x_{11}x_{21} + 16x_{11}x_{25} - 8x_{11}x_{26} - 16x_{11}x_{61} - 5x_{12}^2 \\
&+ 2x_{12}x_{13} + 4x_{12}x_{59} + x_{13}^2 + 7x_{21} - 8x_{21}x_{25} \\
&+ 8x_{21}x_{61} + 2x_{25} + 8x_{25}x_{26} - 4x_{26} + 4x_{26}^2 \\
&- x_{29} + x_{33} - x_{41} + 4x_{59}^2 - x_{60} \\
&- 4x_{61} + 4x_{61}^2 + 6x_{73}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{Q}_{19} = \; &- 1 - x_1 + x_8x_9 - 5x_{10} - x_{10}^2 \\
&- 2x_{10}x_{12} + x_{11} + x_{11}^2 + 2x_{11}x_{12} - 2x_{12} \\
&- x_{25} - x_{26} - 4x_{33} - 4x_{33}^2 + x_{34}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{Q}_{20} = \; &- 1 + x_0 + x_3x_{11} + 18x_{10} + 12x_{10}^2 \\
&+ 11x_{10}x_{11} - 5x_{10}x_{12} - 8x_{10}x_{21} + 16x_{10}x_{22} + 8x_{10}x_{25} \\
&+ 8x_{10}x_{26} + 8x_{10}x_{29} - 4x_{10}x_{41} + 5x_{11} - 12x_{11}^2 \\
&+ 5x_{11}x_{12} - 8x_{11}x_{21} + 64x_{11}x_{22} + 16x_{11}x_{25} + 24x_{11}x_{26} \\
&+ 16x_{11}x_{29} - 14x_{12} + 6x_{12}^2 + 4x_{12}x_{21} + 8x_{12}x_{26} \\
&- 4x_{12}x_{29} - 8x_{12}x_{41} + 17x_{21} - 20x_{21}^2 + 32x_{21}x_{22} \\
&- 8x_{21}x_{25} + 16x_{21}x_{26} + 8x_{21}x_{29} + 2x_{22} - 16x_{22}^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 3x_{23} - 2x_{24} - 3x_{25} + 4x_{25}^2 + x_{26} \\
&+ 4x_{27}^2 - 3x_{28} - 5x_{29} - 3x_{30} - 2x_{32} \\
&- 3x_{33} - 4x_{33}^2 - x_{34} + x_{35} - 4x_{36} \\
&- 2x_{37} + x_{38} - 2x_{40} + 4x_{41} - 4x_{41}^2 \\
&+ x_{42} - 2x_{43} + x_{44} - x_{61} - 4x_{62} \\
&- 2x_{63} - 5x_{73} - x_{77} - x_{204}x_{217} - x_{208}^2 \\
&+ x_{212}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{Q}_{21} = 2 &+ x_0 + x_1 - x_2x_7 - x_3x_{10} \\
&- x_3x_{11} - x_4 + x_5x_{20} - 7x_{10} - 24x_{10}^2 \\
&+ 7x_{10}x_{11} + 5x_{10}x_{12} + 8x_{10}x_{25} - 24x_{10}x_{35} - x_{11} \\
&+ x_{11}^2 - 5x_{11}x_{12} - 2x_{12}^2 - x_{18} - 5x_{21} \\
&- 4x_{25} - 3x_{26} + 4x_{27}^2 - x_{28} + 4x_{33} \\
&+ 4x_{33}^2 - x_{34} + 2x_{35} - 4x_{35}^2 + x_{36} \\
&+ 2x_{45} + 3x_{73} + x_{77} + x_{78} - x_{208}^2
\end{aligned}
$$

$$
\mathcal{Q}_{22} = - x_{11}^2 - 4x_{11}x_{36} - 4x_{36}^2 + x_{37}
$$

$$
\begin{aligned}
\mathcal{Q}_{23} = &- 1 - x_1 + x_2x_7 + x_3x_{10} + x_3x_{11} \\
&+ x_4 - x_5x_{20} - x_{10} - x_{10}^2 - 2x_{10}x_{11} \\
&- 2x_{11} + 2x_{11}x_{12} + x_{12} + x_{12}^2 + x_{18} \\
&- 9x_{21} - 16x_{21}^2 + 2x_{24} - x_{25} + 2x_{28} \\
&+ 2x_{30} + 2x_{32} + 2x_{34} + 4x_{36} + 2x_{37} \\
&+ x_{38} + 2x_{40} + 2x_{43} + 4x_{62} + 2x_{63} \\
&- x_{77} - x_{78} + x_{204}x_{217} - x_{212}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{Q}_{24} = 1 &+ x_0 + x_{10} - x_{10}^2 + 5x_{10}x_{11} \\
&+ 5x_{10}x_{12} + 8x_{10}x_{26} + 18x_{11} - 13x_{11}^2 - 7x_{11}x_{12} \\
&- 16x_{11}x_{25} + 8x_{11}x_{26} - 24x_{11}x_{38} + 7x_{12} + x_{12}^2 \\
&- 8x_{12}x_{25} - 16x_{12}x_{38} - 6x_{21} + 5x_{25} - 12x_{25}^2 \\
&- 8x_{25}x_{26} - 8x_{25}x_{38} + 7x_{26} - 4x_{26}^2 - 4x_{27}^2 \\
&+ x_{28} + x_{29} - 4x_{33} - 4x_{33}^2 + x_{34} \\
&+ 4x_{38} - 4x_{38}^2 + x_{39} - x_{45} - x_{73} \\
&- x_{208}^2
\end{aligned}
$$

$$
\mathcal{Q}_{25} = - x_{12}^2 - 2x_{12}x_{13} - x_{13}^2 + x_{41}
$$

$$
\begin{aligned}
\mathcal{Q}_{26} = &- 1 - x_1 + x_8x_9 - 4x_{10} + 5x_{10}^2 \\
&+ 2x_{10}x_{11} - 6x_{10}x_{12} - 8x_{10}x_{26} - 4x_{10}x_{41} + 5x_{11} \\
&+ x_{11}^2 - 8x_{11}x_{25} - 8x_{11}x_{26} - 13x_{12} + 3x_{12}^2 \\
&+ 16x_{12}x_{26} - 8x_{12}x_{41} + 6x_{21} - 11x_{25} + 8x_{25}^2 \\
&+ 12x_{26} + 4x_{27}^2 - x_{28} - x_{29} + 4x_{33} \\
&+ 4x_{33}^2 - x_{34} + 4x_{41} - 4x_{41}^2 + x_{42} \\
&- 6x_{73}
\end{aligned}
$$

$$\begin{aligned}\mathcal{Q}_{27} = 1 &+ x_1 - x_3x_{11} - x_8x_9 + x_{10}^2\\ &- x_{10}x_{11} - x_{10}x_{12} - 8x_{10}x_{41} - x_{11} - x_{11}^2\\ &- x_{11}x_{12} + x_{12} - 2x_{12}^2 + 2x_{12}x_{13} - 16x_{12}x_{41}\\ &+ x_{13}^2 + x_{21} - 4x_{26} - 16x_{26}^2 - 32x_{26}x_{41}\\ &- 16x_{26}x_{42} + 3x_{41} - 24x_{41}^2 - 16x_{41}x_{42} - 4x_{42}^2\\ &+ x_{43} - x_{73} + x_{77}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{28} = 1 &+ x_0 + x_1 - x_8x_9 - x_{10}\\ &- 10x_{10}^2 + 6x_{10}x_{11} + 2x_{10}x_{12} + 24x_{10}x_{26} + 16x_{10}x_{29}\\ &+ 5x_{11} - 6x_{11}x_{12} - 8x_{11}x_{25} - 8x_{11}x_{26} + x_{12}\\ &- 6x_{12}^2 + 2x_{12}x_{13} + 8x_{12}x_{25} - 8x_{12}x_{26} - 4x_{12}x_{29}\\ &+ x_{13}^2 + x_{21} - 5x_{25} + 8x_{25}x_{26} - 7x_{26}\\ &- 12x_{26}^2 - 16x_{26}x_{29} + 4x_{27}^2 - x_{28} - x_{29}\\ &- 4x_{29}^2 + 4x_{33} + 4x_{33}^2 - x_{34} - x_{41}\\ &+ x_{44} + 3x_{45} - x_{208}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{29} = 1 &+ x_1 - x_8x_9 - x_{10}^2 - 2x_{10}x_{12}\\ &+ x_{11}^2 + 2x_{11}x_{12} - 4x_{12} + 8x_{12}x_{25} + x_{21}\\ &+ 6x_{25} - 4x_{25}^2 + x_{45} - 2x_{73}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{30} = &- 1 - x_2 - x_6 + x_6^2 - x_{10}\\ &+ 2x_{10}^2 - x_{10}x_{11} + 5x_{10}x_{12} - x_{11} - 3x_{11}^2\\ &- 5x_{11}x_{12} + 8x_{12} - 6x_{12}^2 + 2x_{12}x_{13} + 8x_{12}x_{25}\\ &+ 12x_{12}x_{45} + x_{13}^2 - x_{21} - 12x_{25} - 8x_{25}^2\\ &- 16x_{25}x_{45} - x_{26} - x_{41} - 2x_{45} - 4x_{45}^2\\ &+ x_{46} + 3x_{73} - x_{138}x_{192} - x_{192} + x_{193}\\ &- x_{209}^2 - x_{210}^2 - x_{211}^2 - x_{212}^2 + x_{218}^2\\ &+ x_{219}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{31} = 1 &+ x_1 - x_3x_{11} - x_8x_9 + 2x_{11}\\ &- 2x_{11}x_{12} - 4x_{11}x_{46} + x_{12} - 3x_{12}^2 - 8x_{12}x_{46}\\ &- x_{25} - 4x_{46}^2 + x_{47} + x_{77}\end{aligned}$$

$$\mathcal{Q}_{32} = - 1 - 4x_{47} - 4x_{47}^2 + x_{48}$$

$$\begin{aligned}\mathcal{Q}_{33} = 1 &- x_{12}^2 - 8x_{12}x_{47} - 4x_{12}x_{48} + 8x_{47}\\ &- 8x_{47}^2 - 16x_{47}x_{48} - 2x_{48} - 4x_{48}^2 + x_{49}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{34} = 2 &+ x_1 - x_3x_{11} - x_8x_9 + 2x_{11}\\ &- 2x_{11}x_{12} - 4x_{11}x_{49} + x_{12} - 3x_{12}^2 - 8x_{12}x_{49}\\ &- x_{25} - 4x_{49}^2 + x_{50} + x_{77}\end{aligned}$$

$$\mathcal{Q}_{35} = -4x_{44}^2 - 8x_{44}x_{50} - 4x_{50}^2 + x_{51}$$

$$\begin{aligned}\mathcal{Q}_{36} = & -2 - x_1 + x_2x_7 + x_3x_{10} + x_3x_{11} \\ & + x_4 - x_5x_{20} + 6x_{10} + 2x_{10}x_{12} - 8x_{10}x_{44} \\ & - 4x_{10}x_{51} + x_{11} + x_{11}^2 + 2x_{11}x_{12} + 2x_{12}^2 \\ & + x_{18} - x_{25} - x_{26} + 24x_{44} - 12x_{44}^2 \\ & + 8x_{44}x_{50} - 16x_{44}x_{51} + 4x_{50}^2 + 11x_{51} - 4x_{51}^2 \\ & + x_{52} - x_{77} - x_{78}\end{aligned}$$

$$\mathcal{Q}_{37} = -1 - 4x_{52} - 4x_{52}^2 + x_{53}$$

$$\begin{aligned}\mathcal{Q}_{38} = & 1 + x_1 - x_8x_9 - x_{12}^2 - 8x_{12}x_{52} \\ & - 4x_{12}x_{53} + 8x_{52} - 8x_{52}^2 - 16x_{52}x_{53} - 2x_{53} \\ & - 4x_{53}^2 + x_{54}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{39} = & 1 + x_1 - x_8x_9 - 2x_{10} - x_{10}^2 \\ & - x_{10}x_{11} - 3x_{10}x_{12} - 4x_{10}x_{54} + x_{11}x_{12} - 3x_{12} \\ & - x_{12}^2 - 8x_{12}x_{54} + x_{21} - x_{25} - x_{26} \\ & - 4x_{54} - 4x_{54}^2 + x_{55} - x_{73}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{40} = & 2 - x_0 + x_1 - x_2x_7 - x_3x_{10} \\ & - x_3x_{11} - x_4 + x_5x_{20} - x_{18} - 12x_{55} \\ & - 4x_{55}^2 + x_{56} + x_{77} + x_{78} + x_{208}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{41} = & -x_{10}x_{11} + x_{10}x_{12} - x_{11}^2 - x_{11}x_{12} + 2x_{12} \\ & - 2x_{12}^2 + 16x_{12}x_{55} + 8x_{12}x_{56} - 3x_{25} - 16x_{25}x_{55} \\ & - 8x_{25}x_{56} - x_{45} + 8x_{55} - 16x_{55}^2 - 16x_{55}x_{56} \\ & + 4x_{56} - 4x_{56}^2 + x_{57} + x_{73}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{42} = & 1 - x_0 + x_3 - x_3x_{10} - x_3x_{11} \\ & - x_{10} + x_{10}^2 + 2x_{10}x_{12} - 2x_{11} - 4x_{11}^2 \\ & + 2x_{11}x_{12} - 8x_{11}x_{25} - 4x_{11}x_{57} + 4x_{12} - 4x_{12}^2 \\ & + 2x_{12}x_{13} - 12x_{12}x_{45} + 16x_{12}x_{57} + x_{13}^2 - 5x_{25} \\ & + 4x_{25}^2 + 16x_{25}x_{45} - 16x_{25}x_{57} - x_{26} - x_{41} \\ & - x_{45} + 4x_{45}^2 - x_{46} - 12x_{50} - 4x_{50}^2 \\ & - 4x_{57}^2 + x_{58} + x_{77} + x_{78} + x_{208}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{43} = & -2 + x_0 + x_0x_2 - 2x_1 + x_2x_7 \\ & + x_3^2 + x_3x_{10} + x_3x_{11} - x_5x_{20} - x_6 \\ & + x_6^2 + x_8x_9 - x_9 + x_{18} + 24x_{50} \\ & - 8x_{50}^2 - 16x_{50}x_{58} - 2x_{58} - 4x_{58}^2 + x_{59} \\ & - x_{77} - x_{78} - x_{138}x_{192} - x_{192} + x_{193} \\ & - 2x_{208}^2 - x_{209}^2 - x_{210}^2 - x_{211}^2 - x_{212}^2 \\ & + x_{218}^2 + x_{219}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{44} = & 2 + x_1 - x_2x_7 - x_3x_{10} - x_3x_{11} \\ & - x_4 + x_5x_{20} - 7x_{10} + x_{10}^2 - x_{10}x_{11} \\ & - x_{10}x_{12} + 8x_{10}x_{61} - 9x_{11} + 4x_{11}^2 + x_{11}x_{12} \\ & + 16x_{11}x_{61} - 3x_{12}^2 + 2x_{12}x_{13} + 4x_{12}x_{21} - 4x_{12}x_{59} \\ & + x_{13}^2 - x_{18} + 6x_{21} - 8x_{21}x_{61} - 2x_{25} \\ & + x_{26} - x_{41} - 4x_{59}^2 + x_{60} + 3x_{61} \\ & - 4x_{61}^2 + 5x_{73} + x_{77} + x_{78}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{45} = & 2 + x_1 - x_3x_{11} - x_8x_9 + 4x_{10} \\ & + 2x_{10}^2 + 8x_{10}x_{21} + 3x_{11} - 11x_{11}^2 + 2x_{11}x_{12} \\ & + 16x_{11}x_{21} + 2x_{12} + 2x_{12}^2 - 4x_{12}x_{21} - 10x_{21} \\ & - 4x_{21}^2 + x_{25} + x_{61} - 4x_{73} + x_{77}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{46} = & 1 + x_1 - x_8x_9 + 2x_{10} + x_{10}^2 \\ & - 3x_{10}x_{11} - 3x_{10}x_{12} + 8x_{10}x_{21} - 8x_{10}x_{25} + 16x_{10}x_{60} \\ & - 3x_{11} + x_{11}^2 - 3x_{11}x_{12} + 16x_{11}x_{21} + 32x_{11}x_{60} \\ & - 6x_{12} + 5x_{12}^2 - 2x_{12}x_{13} + 12x_{12}x_{21} + 8x_{12}x_{59} \\ & - x_{13}^2 - x_{21} - 4x_{21}^2 - 16x_{21}x_{60} - 4x_{27}^2 \\ & + x_{28} - 4x_{33} - 4x_{33}^2 + x_{34} + x_{41} \\ & - 2x_{45} + 8x_{59}^2 - 2x_{60} - 4x_{60}^2 - 3x_{61} \\ & + x_{62} + 5x_{73}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{47} = & -1 - x_1 + x_8x_9 - 4x_{62} - 4x_{62}^2 \\ & + x_{63}\end{aligned}$$

$$\mathcal{Q}_{48} = -x_3x_{15} + x_{64}$$

$$\mathcal{Q}_{49} = x_{14} - x_{15} - 2x_{24}x_{64} - x_{65}$$

$$\mathcal{Q}_{50} = x_{14} - x_{15} - 2x_{28}x_{64} - x_{66}$$

$$\mathcal{Q}_{51} = -x_3x_{15} - 2x_{30}x_{64} + 2x_{43}x_{64} - x_{64} - x_{67} + x_{72}$$

$$\mathcal{Q}_{52} = x_{14} - x_{15} - 2x_{32}x_{64} - x_{68}$$

$$\mathcal{Q}_{53} = x_{14} - x_{15} + 2x_{24}x_{64} - 2x_{30}x_{64} - 2x_{34}x_{64} + x_{65} - x_{67} - x_{69}$$

$$\mathcal{Q}_{54} = x_{14} - x_{15} - 4x_{36}x_{64} - 2x_{37}x_{64} - x_{70}$$

$$\mathcal{Q}_{55} = x_{14} - x_{15} - 2x_{40}x_{64} - x_{71}$$

$$\mathcal{Q}_{56} = -2x_3x_{15} + x_{14} - x_{15} - 2x_{43}x_{64} - 2x_{64} - x_{72}$$

$$\mathcal{Q}_{57} = -x_{10}x_{11} + x_{10}x_{12} + x_{11} + x_{11}x_{12} - x_{12} - x_{12}^2 + x_{73}$$

$$\mathcal{Q}_{58} = 1 + 2x_{13} + x_{13}^2 - x_{73}^2 + x_{74}$$

$$\mathcal{Q}_{59} = x_3x_{15} - 4x_{62}x_{64} - 2x_{63}x_{64} + x_{64} + x_{75}$$

$$\mathcal{Q}_{60} = -x_{14}^2 + 2x_{14}x_{15} + 2x_{14}x_{75} - x_{15}^2 - 2x_{15}x_{75} - x_{74}^2 - x_{75}^2 + x_{76}$$

$$\mathcal{Q}_{61} = -1 - x_3x_{11} + x_{77}$$

$$\mathcal{Q}_{62} = -x_{15}x_{77} + x_{79}x_{80} + x_{81}$$

$$\mathcal{Q}_{63} = -1 + x_3 - x_3x_{10} + x_{78}$$

$$\mathcal{Q}_{64} = -2 - x_2x_7 + x_3 + 2x_3x_{10} + 3x_3x_{11} - 2x_3x_{21} + x_8x_9 + x_{77} + x_{79}$$

$$\mathcal{Q}_{65} = -x_{14}x_{78} + x_{80}$$

$$\begin{aligned}\mathcal{Q}_{66} = {} & 5x_{14} + x_{14}x_{77} + 7x_{15} - 5x_{15}x_{77} - 4x_{15}x_{78} \\ & - x_{15}x_{79} - 2x_{24}x_{64} - 2x_{28}x_{64} - 2x_{32}x_{64} - 2x_{34}x_{64} \\ & - 2x_{40}x_{64} - x_{65} - x_{66} - x_{68} - x_{69} \\ & - x_{71} - 3x_{79}x_{80} - 3x_{81} + x_{82}\end{aligned}$$

$$\mathcal{Q}_{67} = -x_{14}x_{65} + x_{15}x_{65} + x_{83}$$

$$\mathcal{Q}_{68} = -x_{66}x_{83} + x_{84}$$

$$\mathcal{Q}_{69} = -x_{67}x_{84} + x_{85}$$

$$\mathcal{Q}_{70} = -x_{68}x_{85} + x_{86}$$

$$\mathcal{Q}_{71} = -x_{69}x_{86} + x_{87}$$

$$\mathcal{Q}_{72} = -x_{70}x_{87} + x_{88}$$

$$\mathcal{Q}_{73} = -x_{71}x_{88} + x_{89}$$

$$\mathcal{Q}_{74} = -x_{72}x_{89} + x_{90}$$

$$\mathcal{Q}_{75} = -x_{81}x_{90} + x_{91}$$

$$\mathcal{Q}_{76} = -x_{82}x_{91} + x_{92}$$

$$\mathcal{Q}_{77} = -x_{92}^2 + x_{93}$$

$$\mathcal{Q}_{78} = -x_2 - x_2x_{18} + x_{18} + x_{94}$$

$$\mathcal{Q}_{79} = -x_{16}x_{94} + x_{95}$$

$$\begin{aligned}\mathcal{Q}_{80} = & -2x_1^2 + 2x_1x_{14} + 2x_1x_{15} + 2x_1x_{95} + 2x_1x_{96} \\ & - x_{14}^2 - 2x_{14}x_{95} - x_{15}^2 - 2x_{15}x_{96} + x_{19}x_{97} \\ & - x_{76}x_{93} - x_{95}^2 - x_{96}^2\end{aligned}$$

$$\mathcal{Q}_{81} = -x_2x_{17} + x_{17} - x_{17}x_{94} + x_{96}$$

$$\mathcal{Q}_{82} = -x_{100} - 2x_{100}x_{231} + x_{102}$$

$$\mathcal{Q}_{83} = -x_5x_{124} + x_{98}x_{222} - x_{101} - x_{102}x_{124}$$

$$\mathcal{Q}_{84} = x_{103}x_{223} - x_{105}x_{127} - x_{106}$$

$$\mathcal{Q}_{85} = x_{107}x_{223} - x_{108}x_{127} - x_{109}$$

$$\mathcal{Q}_{86} = x_{110}x_{223} - x_{111}x_{127} - x_{112}$$

$$\mathcal{Q}_{87} = x_{113}x_{223} - x_{114}x_{127} - x_{115}$$

$$\mathcal{Q}_{88} = x_{116}x_{224} - x_{118}x_{127} - x_{119}$$

$$\mathcal{Q}_{89} = x_{120}x_{224} - x_{121}x_{127} - x_{122}$$

$$\mathcal{Q}_{90} = -x_{19}x_{124} + x_{127}$$

$$\mathcal{Q}_{91} = x_{123} - x_{125}x_{127} - x_{126}$$

$$\mathcal{Q}_{92} = -x_{127}x_{129} + x_{128} - x_{130}$$

$$\mathcal{Q}_{93} = -x_{127}x_{132} + x_{131}x_{222} - x_{133}$$

$$\mathcal{Q}_{94} = -x_{127}x_{135} + x_{134}x_{223} - x_{136}$$

$$\mathcal{Q}_{95} = x_{137} - 2x_{137}x_{139} + x_{140}$$

$$\mathcal{Q}_{96} = -x_{137}x_{140} - x_{140} + x_{141}$$

$$\mathcal{Q}_{97} = x_{140}^2 - x_{141}^2 + x_{161}$$

$$\mathcal{Q}_{98} = -x_{138}x_{143} - x_{143} + x_{144}$$

$$\mathcal{Q}_{99} = 2x_{142} + x_{142}^2 + x_{143}^2 - x_{144}^2$$

$$\mathcal{Q}_{100} = x_4^2 - 2x_4x_{138} + x_{138}^2 + x_{145} - x_{161}$$

$$\begin{aligned}\mathcal{Q}_{101} = &-1 - x_4x_{143} + x_6 + x_{138}x_{143} - x_{142} \\ &+ x_{145}x_{146}\end{aligned}$$

$$\mathcal{Q}_{102} = 1 - 2x_5 + x_5^2 - 2x_5x_{138} + x_{147}$$

$$\mathcal{Q}_{103} = -x_5x_{192} + x_{19} + x_{147}x_{148} - x_{191} + x_{193}$$

$$\mathcal{Q}_{104} = -1 + 2x_{138}x_{207} - x_{149} + 2x_{207} - x_{207}^2$$

$$\begin{aligned}\mathcal{Q}_{105} = &1 + x_{149}x_{150} - x_{191} - x_{192}x_{207} + x_{193} \\ &+ 2x_{231}\end{aligned}$$

$$\mathcal{Q}_{106} = -x_{138}x_{152} - x_{152} + x_{153}$$

$$\mathcal{Q}_{107} = 2x_{151} + x_{151}^2 + x_{152}^2 - x_{153}^2$$

$$\begin{aligned}\mathcal{Q}_{108} = &-x_{138} + x_{138}^2 - 4x_{138}x_{231} - x_{149}x_{150} + x_{154} \\ &- x_{161} + x_{191} + x_{192}x_{207} - x_{193} + x_{209}x_{215} \\ &- x_{213} + 2x_{231} + 4x_{231}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{109} = &-1 + x_{98} + x_{138}x_{152} - x_{151} - x_{152} \\ &- 2x_{152}x_{231} + x_{154}x_{155}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{110} = & -1 + x_{99} + x_{138}x_{143} - 2x_{138}x_{156} - x_{142} \\ & - 2x_{143}x_{231} + x_{154}x_{156} + x_{156} + 4x_{156}x_{231}\end{aligned}$$

$$\mathcal{Q}_{111} = 1 + x_{138} - x_{138}x_{158} - x_{158} + x_{159}$$

$$\begin{aligned}\mathcal{Q}_{112} = & -x_{138} + x_{138}x_{158} + 2x_{157} + x_{157}^2 - x_{158} \\ & + x_{158}^2 - x_{159} - x_{159}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{113} = & x_{103} - x_{138} + x_{138}x_{158} + x_{154}x_{160} - x_{157} \\ & - x_{158} - 2x_{158}x_{231} + 2x_{231}\end{aligned}$$

$$\mathcal{Q}_{114} = -x_{138} - x_{138}^2 + x_{161} + x_{209}x_{215} - x_{213}$$

$$\begin{aligned}\mathcal{Q}_{115} = & x_{104} + x_{138}x_{144} - 2x_{138}x_{162} - x_{142} - 2x_{142}x_{231} \\ & - x_{143}x_{161} - 2x_{144}x_{231} + x_{149}x_{150} + x_{154}x_{162} + x_{162} \\ & + 4x_{162}x_{231} - x_{191} - x_{192}x_{207} + x_{193}\end{aligned}$$

$$\mathcal{Q}_{116} = -x_{138}x_{164} - x_{164} + x_{165}$$

$$\mathcal{Q}_{117} = 2x_{163} + x_{163}^2 + x_{164}^2 - x_{165}^2$$

$$\begin{aligned}\mathcal{Q}_{118} = & -1 + x_{107} + x_{138}x_{164} + x_{154}x_{166} - x_{163} \\ & - x_{164} - 2x_{164}x_{231}\end{aligned}$$

$$\mathcal{Q}_{119} = -x_{138}x_{168} - x_{168} + x_{169}$$

$$\mathcal{Q}_{120} = 2x_{167} + x_{167}^2 + x_{168}^2 - x_{169}^2$$

$$\begin{aligned}\mathcal{Q}_{121} = & -1 + x_{110} + x_{138}x_{168} + x_{154}x_{170} - x_{167} \\ & - x_{168} - 2x_{168}x_{231}\end{aligned}$$

$$\mathcal{Q}_{122} = -x_{138}x_{172} - x_{172} + x_{173}$$

$$\mathcal{Q}_{123} = 2x_{171} + x_{171}^2 + x_{172}^2 - x_{173}^2$$

$$\begin{aligned}\mathcal{Q}_{124} = & -1 + x_{113} + x_{138}x_{172} + x_{154}x_{174} - x_{171} \\ & - x_{172} - 2x_{172}x_{231}\end{aligned}$$

$$\mathcal{Q}_{125} = -x_{138}x_{176} - x_{176} + x_{177}$$

$$\mathcal{Q}_{126} = 2x_{175} + x_{175}^2 + x_{176}^2 - x_{177}^2$$

$$\begin{aligned}\mathcal{Q}_{127} = & -1 + x_{116} + x_{138}x_{176} + x_{154}x_{178} - x_{175} \\ & - x_{176} - 2x_{176}x_{231}\end{aligned}$$

$$\mathcal{Q}_{128} = -x_{138}x_{180} - x_{180} + x_{181}$$

$$\mathcal{Q}_{129} = 2x_{179} + x_{179}^2 + x_{180}^2 - x_{181}^2$$

$$\begin{aligned}\mathcal{Q}_{130} = & -1 + x_{117} + x_{138}x_{180} - 2x_{138}x_{182} + x_{154}x_{182} \\ & - x_{179} - 2x_{180}x_{231} + x_{182} + 4x_{182}x_{231}\end{aligned}$$

$$\mathcal{Q}_{131} = -x_{138}x_{184} - x_{184} + x_{185}$$

$$\mathcal{Q}_{132} = 2x_{183} + x_{183}^2 + x_{184}^2 - x_{185}^2$$

$$\begin{aligned}\mathcal{Q}_{133} = & -1 + x_{120} + x_{138}x_{184} + x_{154}x_{186} - x_{183} \\ & - x_{184} - 2x_{184}x_{231}\end{aligned}$$

$$\mathcal{Q}_{134} = -x_{138}x_{188} - x_{188} + x_{189}$$

$$\mathcal{Q}_{135} = 2x_{187} + x_{187}^2 + x_{188}^2 - x_{189}^2$$

$$\begin{aligned}\mathcal{Q}_{136} = & -1 + x_{123} + x_{138}x_{188} + x_{154}x_{190} - x_{187} \\ & - x_{188} - 2x_{188}x_{231}\end{aligned}$$

$$\mathcal{Q}_{137} = 1 - x_{138}x_{192} - x_{192} + x_{193}$$

$$\mathcal{Q}_{138} = -1 - x_5x_{192} + x_{19} + x_{147}x_{148} + x_{191} + x_{191}^2 + x_{192}^2 - x_{193} - x_{193}^2$$

$$\mathcal{Q}_{139} = x_{124} - 2x_{138}x_{194} + x_{154}x_{194} - x_{191} - x_{192} - 2x_{192}x_{231} + x_{193} + x_{194} + 4x_{194}x_{231}$$

$$\mathcal{Q}_{140} = -x_{138}x_{196} - x_{196} + x_{197}$$

$$\mathcal{Q}_{141} = 2x_{195} + x_{195}^2 + x_{196}^2 - x_{197}^2$$

$$\mathcal{Q}_{142} = -1 + x_{128} + x_{138}x_{196} + x_{154}x_{198} - x_{195} - x_{196} - 2x_{196}x_{231}$$

$$\mathcal{Q}_{143} = -x_{138}x_{200} - x_{200} + x_{201}$$

$$\mathcal{Q}_{144} = 2x_{199} + x_{199}^2 + x_{200}^2 - x_{201}^2$$

$$\mathcal{Q}_{145} = -1 + x_{131} + x_{138}x_{200} + x_{154}x_{202} - x_{199} - x_{200} - 2x_{200}x_{231}$$

$$\mathcal{Q}_{146} = -x_{138}x_{204} - x_{204} + x_{205}$$

$$\mathcal{Q}_{147} = 2x_{203} + x_{203}^2 + x_{204}^2 - x_{205}^2$$

$$\mathcal{Q}_{148} = -1 + x_{134} + x_{138}x_{204} + x_{154}x_{206} - x_{203} - x_{204} - 2x_{204}x_{231}$$

$$\begin{aligned}\mathcal{Q}_{149} = & -1-2x_0-3x_0^2-2x_0x_4-x_1\\ & -x_1^2-x_2-x_2^2-x_4-2x_4^2\\ & -x_5^2-x_5x_{20}+x_5x_{192}-x_6+x_8x_9\\ & +x_{10}+2x_{10}x_{12}-x_{11}^2-x_{12}-x_{13}^2\\ & -x_{14}^2-x_{15}^2-x_{16}^2-x_{17}^2+x_{18}\\ & -x_{18}^2-x_{19}-x_{19}^2-x_{26}-x_{98}^2\\ & -x_{99}^2-x_{103}^2-x_{104}^2-x_{107}^2-x_{110}^2\\ & -x_{113}^2-x_{116}^2-x_{117}^2-x_{120}^2-x_{123}^2\\ & -x_{124}^2-x_{128}^2-x_{131}^2-x_{134}^2+x_{137}\\ & -x_{138}+x_{138}^2-4x_{138}x_{231}+x_{143}^2-x_{147}x_{148}\\ & +2x_{149}x_{150}+x_{152}^2+x_{154}-2x_{158}+x_{158}^2\\ & -x_{161}+x_{164}^2+x_{168}^2+x_{172}^2+x_{176}^2\\ & +x_{180}^2+x_{184}^2+x_{188}^2-x_{191}+x_{192}^2\\ & -2x_{192}x_{207}+x_{193}+x_{196}^2+x_{200}^2-x_{204}^2\\ & -x_{207}^2+x_{209}x_{215}-x_{213}+x_{225}^2+x_{226}^2\\ & +x_{227}+x_{227}^2-4x_{231}-4x_{231}^2\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{150} = & -x_5x_{10}-x_5x_{11}-x_5x_{12}-x_5x_{13}-x_5x_{14}\\ & -x_5x_{15}-x_5x_{16}-x_5x_{17}-x_5x_{18}+x_{207}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{151} = & -2+x_0-2x_1-x_2+2x_2x_7\\ & -x_3-x_3x_{10}-x_3x_{11}+2x_3x_{21}+x_4\\ & -x_5x_{20}-x_6+x_6^2+x_{11}+x_{11}^2\\ & +2x_{11}x_{12}+x_{12}+x_{12}^2+x_{18}-x_{25}\\ & -3x_{77}-x_{78}-x_{79}-x_{138}x_{192}-x_{192}\\ & +x_{193}-x_{208}^2-x_{209}^2-x_{210}^2-x_{211}^2\\ & -x_{212}^2+x_{218}^2+x_{219}^2\end{aligned}$$

$$\mathcal{Q}_{152} = -1+x_2+x_6-x_6^2$$

$$\mathcal{Q}_{153} = -x_{138}x_{210}-x_{210}+x_{218}$$

$$\mathcal{Q}_{154} = -1+x_{209}^2+x_{210}^2-x_{218}^2$$

$$\mathcal{Q}_{155} = -x_{212}-x_{212}x_{213}+x_{219}$$

$$\mathcal{Q}_{156} = -1+x_{211}^2+x_{212}^2-x_{219}^2$$

$$\mathcal{Q}_{157} = -x_{204}^2+x_{220}$$

$$\mathcal{Q}_{158} = x_{210} - 2x_{214}x_{220} - x_{220}$$

$$\mathcal{Q}_{159} = x_{138}x_{209} + x_{215}$$

$$\mathcal{Q}_{160} = -x_{138} - x_{209}x_{215} + x_{213}$$

$$\mathcal{Q}_{161} = -1 - x_{203} - x_{209}x_{216} + x_{211}$$

$$\begin{aligned}\mathcal{Q}_{162} = & -x_{10} - x_{11} - 4x_{21} - 2x_{24} - 2x_{28} \\ & - 2x_{30} - 2x_{32} - 2x_{34} - 4x_{36} - 2x_{37} \\ & - 2x_{40} - 2x_{43} - 4x_{62} - 2x_{63} - x_{204}x_{217} \\ & + x_{212}\end{aligned}$$

$$\begin{aligned}\mathcal{Q}_{163} = & -1 + 2x_0^2 + 2x_0x_4 - 2x_0x_{143} - 2x_0x_{152} \\ & + x_1^2 - 2x_1x_{192} + x_2^2 - 2x_2x_{180} + x_4^2 \\ & - 2x_4x_{152} + x_{10}^2 - 2x_{10}x_{158} - 2x_{10}x_{204} + x_{11}^2 \\ & - 2x_{11}x_{164} - 2x_{11}x_{204} + x_{12}^2 - 2x_{12}x_{168} + x_{13}^2 \\ & - 2x_{13}x_{172} + x_{14}^2 - 2x_{14}x_{176} + x_{15}^2 - 2x_{15}x_{184} \\ & + x_{16}^2 - 2x_{16}x_{188} + x_{17}^2 - 2x_{17}x_{196} + x_{18}^2 \\ & - 2x_{18}x_{200} - 8x_{21}x_{204} - 4x_{24}x_{204} - 4x_{28}x_{204} - 4x_{30}x_{204} \\ & - 4x_{32}x_{204} - 4x_{34}x_{204} - 8x_{36}x_{204} - 4x_{37}x_{204} - 4x_{40}x_{204} \\ & - 4x_{43}x_{204} - 8x_{62}x_{204} - 4x_{63}x_{204} + x_{138} - x_{138}x_{158} \\ & - x_{138}x_{221} + x_{142}^2 + x_{151}^2 + x_{157}^2 + x_{158} \\ & + x_{159} + x_{163}^2 + x_{167}^2 + x_{171}^2 + x_{175}^2 \\ & + x_{179}^2 + x_{183}^2 + x_{187}^2 + x_{191}^2 + x_{195}^2 \\ & + x_{199}^2 + x_{203}^2 + x_{204}^2 + x_{220} - x_{225}^2 \\ & - x_{226}^2 - x_{227} - x_{227}^2\end{aligned}$$

$$\mathcal{Q}_{164} = -x_{99}x_{222} + x_{124}$$

$$\mathcal{Q}_{165} = -x_{104}x_{223} + x_{124}$$

$$\mathcal{Q}_{166} = -x_{117}x_{224} + x_{124}$$

$$
\begin{aligned}
\mathcal{Q}_{167} = {} & 2 + 2x_1 - x_2x_7 - x_3x_{10} - x_3x_{11} \\
& - x_4 + x_5x_{20} - x_8x_9 + x_{10} + x_{10}^2 \\
& + 2x_{10}x_{11} - 8x_{10}x_{21} - 4x_{10}x_{24} - 4x_{10}x_{28} - 4x_{10}x_{30} \\
& - 4x_{10}x_{32} - 4x_{10}x_{34} - 8x_{10}x_{36} - 4x_{10}x_{37} - 4x_{10}x_{40} \\
& - 4x_{10}x_{43} - 8x_{10}x_{62} - 4x_{10}x_{63} + 2x_{11} + 3x_{11}^2 \\
& - 2x_{11}x_{12} - 8x_{11}x_{21} - 4x_{11}x_{24} - 4x_{11}x_{28} - 4x_{11}x_{30} \\
& - 4x_{11}x_{32} - 4x_{11}x_{34} + 4x_{11}x_{36} - 4x_{11}x_{37} - 4x_{11}x_{40} \\
& - 4x_{11}x_{43} - 8x_{11}x_{62} - 4x_{11}x_{63} - x_{12} - x_{12}^2 \\
& - x_{18} + 8x_{21} - 16x_{21}x_{24} - 16x_{21}x_{28} - 16x_{21}x_{30} \\
& - 16x_{21}x_{32} - 16x_{21}x_{34} - 32x_{21}x_{36} - 16x_{21}x_{37} - 16x_{21}x_{40} \\
& - 16x_{21}x_{43} - 32x_{21}x_{62} - 16x_{21}x_{63} - 2x_{24} - 4x_{24}^2 \\
& - 8x_{24}x_{28} - 8x_{24}x_{30} - 8x_{24}x_{32} - 8x_{24}x_{34} - 16x_{24}x_{36} \\
& - 8x_{24}x_{37} - 8x_{24}x_{40} - 8x_{24}x_{43} - 16x_{24}x_{62} - 8x_{24}x_{63} \\
& + x_{25} - 2x_{28} - 4x_{28}^2 - 8x_{28}x_{30} - 8x_{28}x_{32} \\
& - 8x_{28}x_{34} - 16x_{28}x_{36} - 8x_{28}x_{37} - 8x_{28}x_{40} - 8x_{28}x_{43} \\
& - 16x_{28}x_{62} - 8x_{28}x_{63} - 2x_{30} - 4x_{30}^2 - 8x_{30}x_{32} \\
& - 8x_{30}x_{34} - 16x_{30}x_{36} - 8x_{30}x_{37} - 8x_{30}x_{40} - 8x_{30}x_{43} \\
& - 16x_{30}x_{62} - 8x_{30}x_{63} - 2x_{32} - 4x_{32}^2 - 8x_{32}x_{34} \\
& - 16x_{32}x_{36} - 8x_{32}x_{37} - 8x_{32}x_{40} - 8x_{32}x_{43} - 16x_{32}x_{62} \\
& - 8x_{32}x_{63} - 2x_{34} - 4x_{34}^2 - 16x_{34}x_{36} - 8x_{34}x_{37} \\
& - 8x_{34}x_{40} - 8x_{34}x_{43} - 16x_{34}x_{62} - 8x_{34}x_{63} - 4x_{36} \\
& - 4x_{36}^2 - 16x_{36}x_{37} - 16x_{36}x_{40} - 16x_{36}x_{43} - 32x_{36}x_{62} \\
& - 16x_{36}x_{63} - 5x_{37} - 4x_{37}^2 - 8x_{37}x_{40} - 8x_{37}x_{43} \\
& - 16x_{37}x_{62} - 8x_{37}x_{63} - x_{38} - 2x_{40} - 4x_{40}^2 \\
& - 8x_{40}x_{43} - 16x_{40}x_{62} - 8x_{40}x_{63} - 2x_{43} - 4x_{43}^2 \\
& - 16x_{43}x_{62} - 8x_{43}x_{63} + 4x_{62} - 8x_{62}^2 - 16x_{62}x_{63} \\
& - 4x_{63} - 4x_{63}^2 + x_{77} + x_{78} + x_{138}x_{192} \\
& - x_{143}^2 - x_{152}^2 + 2x_{158} - x_{158}^2 - x_{164}^2 \\
& - x_{168}^2 - x_{172}^2 - x_{176}^2 - x_{180}^2 - x_{184}^2 \\
& - x_{188}^2 + x_{192} - x_{192}^2 - x_{193} - x_{196}^2 \\
& - x_{200}^2 + x_{204}^2 - x_{204}x_{217} + x_{212} - x_{225}^2 \\
& - x_{226}^2 - x_{227} - x_{227}^2
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{Q}_{168} = {} & - 1 - 4x_{101}^2 - 4x_{106}^2 - 4x_{109}^2 - 4x_{112}^2 \\
& - 4x_{115}^2 - 4x_{119}^2 - 4x_{122}^2 + x_{124}^2 - 4x_{126}^2 \\
& - 4x_{130}^2 - 4x_{133}^2 - 4x_{136}^2 - x_{228}^2 - x_{229}^2 \\
& - x_{230} - x_{230}^2 - 4x_{233}^2
\end{aligned}
$$

$$
\mathcal{Q}_{169} = - x_5x_{232} + x_{124}
$$

$$
\mathcal{Q}_{170} = x_3x_{124} - x_6x_{232} - x_{233}
$$